Where Is the Decision Made? Probe-Weighted Token Credit for Critic-Free Agent Reinforcement Learning

Anonymous CVPR submission

Abstract

Critic-free reinforcement learning has become the standard way to train language-model agents, but it assigns credit coarsely: GRPO gives every token of a trajectory the same advantage, and GiGPO and GACA refine this to the level of an environment step. Inside a step, however, a reasoning model writes a hundred tokens of chain of thought before a ten-token action, and all of them still share one advantage. We ask where in that response the decision is actually made. A counterfactual flip test on ALFWorld (truncate the reasoning, resample the rest, and check whether the action changes) shows that 82% of the decision is settled before the action is written, that token entropy is a poor guide to where (the final tenth of a response holds 19% of the decision but 3% of the entropy), and that a cheap teacher-forced commitment probe*, the measurement behind Causal Importance of Reasoning, tracks it closely: weighting tokens by the probe's rise places* $1.65\times$ *more credit on decisive segments than uniform credit, consistently across task types, held-out games and a trained policy, whereas the model's own attention from action to reasoning does so only weakly* ($1.12\times$)*. We turn the probe into a mean-preserving, clipped redistribution of each step's GACA advantage over its tokens. In a budget-matched study on ALFWorld with Qwen2.5-1.5B-Instruct and full fine-tuning (about 6% of GACA's training budget, same 128 held-out games for every method), probe-weighted credit reaches the highest final success of all arms (20.3% against 18.8% for GACA and 14.1% for GRPO), but none of the method differences is statistically resolved at this budget, and a variant that applies the probe to positive advantages only performs significantly worse than both baselines* ***[TODO: frozen-reference result]****. We also find that GACA's per-step criticality score is about 85% within-state noise. Code, logs and weights are released.*

1. Introduction

Reinforcement learning from outcome rewards has become the standard recipe for training language models to reason [6, 19] and, increasingly, to act as agents over many turns [5, 11, 25]. The dominant algorithms are *critic-free*: instead of learning a value function, GRPO [19] samples a group of trajectories for the same prompt and standardizes each trajectory's return against the group. The price is coarse credit. In a fifty-step household task every token of every step receives the same advantage, whether it was the decisive choice of where to look or a routine move along the way.

Two recent methods refine this to the level of an environment step. GiGPO [5] regroups the visits that different rollouts make to the same environment state (an *anchor state*) and contrasts their discounted returns, which yields a step-level advantage without a critic. GACA [11] observes that the right granularity depends on the state and mixes the episode- and step-level advantages with a per-step weight λ_t set by the policy's uncertainty, the mean token negative log-likelihood (NLL) of the sampled action; it reports 96.9% success on ALFWorld at 1.5B parameters, against 86.7% for GiGPO and 72.8% for GRPO.

Both methods stop at the step. Yet a modern agent's "action" is not a short command: in the reasoning format these methods are trained with, the model writes roughly a hundred tokens of chain of thought inside `<think>` tags and then a ten-token command inside `<action>` tags, and every one of those tokens still receives its step's single advantage. If a step was good because of one sentence of planning, the policy gradient spends most of its credit reinforcing the boilerplate around it. This paper asks a direct question: *where inside a response is the decision actually made, and can credit be sent there?*

We answer it in two parts. First, **measurement**. We run a counterfactual flip test on real ALFWorld steps: cut the response after j tokens, resample the remainder many times, and record how often the final action changes. The drop of that flip rate across a segment is the share of the decision settled inside it. This yields three findings. (i) The decision is made in the reasoning: 82% of it is settled before the `<action>` tag. (ii) Token entropy, the signal behind recent token-level methods for single-turn reasoning [24], does not locate it: the last tenth of a response carries 19% of the decision but only 3% of the entropy, and entropy weighting moves credit toward decisive segments by only 4%. (iii) A cheap *commitment probe* does: at each cut, force the reasoning closed and score the probability of the eventual

action under the model. This is exactly the truncate-and-force measurement that Yu *et al*. [28] use to define Causal Importance of Reasoning (CIR) and that Lanham *et al*. [10] call early answering. Weighting each token by the probe's rise across its segment places $1.65\times$ more credit on decisive segments than uniform credit at our training configuration, with the effect consistent across all six task types, on held-out games, and on a trained policy. Along the way we find that GACA's own per-step score is about 85% noise from which reasoning path happened to be sampled.

Second, **training**. Yu *et al*. use CIR as a scalar bonus in the reward; every token of a response still shares one advantage. We instead use the probe's per-segment rise to *redistribute* each step's GACA advantage over the tokens of its response, with weights that are clipped and average to one, so a step's total credit is unchanged and only where it lands moves (Fig. 1). The probe costs a teacher-forced scoring pass and no generation. In a study on ALFWorld with full fine-tuning of Qwen2.5-1.5B-Instruct inside the authors' own training stack (verl-agent [5, 20]), with every method evaluated on the same 128 held-out games and compared game by game, probe-weighted credit reaches the highest final success of any arm. Our budget is about 6% of GACA's, one seed per arm; at that budget no difference between methods is statistically resolved, and we report it as such.

Contributions.

- A counterfactual measurement of where decisions are made inside agent reasoning, and evidence that the commitment probe tracks it while entropy and NLL do not and attention only weakly does (Sec. 6).
- Probe-weighted token credit: a critic-free redistribution of step-level advantage over tokens, compatible with GRPO, GiGPO and GACA (Sec. 4).
- A budget-matched, game-paired comparison of GRPO, GACA and five treatments on ALFWorld, with negative results reported (Sec. 7), and a measurement that GACA's criticality score is dominated by within-state noise.

2. Related Work

Critic-free RL for language models and agents. PPO [18] with a learned value function [17] was the original workhorse of RL from human feedback [15]. GRPO [19] removed the critic by standardizing each sampled response's reward against a group of responses to the same prompt, and with its successors (REINFORCE-style estimators [1], DAPO's clipping and sampling changes [27]) it became the default for reasoning models [6]. Carrying it to multi-turn agents required new infrastructure (HybridFlow/verl [20], RAGEN [25]); an alternative line keeps a critic but makes it hierarchical across turns [29].

Step-level credit without a critic. GiGPO [5] is, to our knowledge, the first critic-free step-level advantage for LLM agents. It hashes environment states across the group's rollouts into *anchor states* and standardizes the discounted return of each visit against the other visits to the same state, then adds this step advantage to the episode one. GACA [11] argues that the useful granularity is a property of the state and mixes the two levels with a per-step weight λ_t driven by a criticality score, the sampled action's mean token NLL normalized within its trajectory; it proves that a state-adaptive weight dominates any constant one and that the optimal weight grows with the anchor-group size. Its Appendix B.4 proposes, but does not train, a leave-one-out score that averages the NLL of the other visits to the same anchor. Both methods stop at the environment step: all tokens of a step's response share one advantage. Process reward models [12, 23] and Monte-Carlo value estimates from intermediate states [8] offer finer credit at the cost of a learned verifier or many extra rollouts; reward redistribution methods such as RUDDER [2] reshape returns, which reward-shaping theory constrains [14]. Our method adds no model and no rollouts.

Token-level signals in RL for reasoning. Wang *et al*. [24] show that in single-turn reasoning a small minority of high-entropy "forking" tokens drives most of the improvement, and that restricting the policy gradient to them helps; Cui *et al*. [4] study how RL collapses policy entropy. GACA also scores steps by uncertainty. We test whether uncertainty locates the *decision* inside an agent's reasoning and find that it largely does not (Sec. 6).

Faithfulness and causal importance of reasoning. Lanham *et al*. [10] test whether chain of thought is used by truncating it and forcing an early answer. Bogdan *et al*. [3] locate "thought anchors", the sentences that shape a reasoning trace, by resampling the trace from each sentence and, as a white-box alternative, by inspecting attention between sentences. Yu *et al*. [28] formalize the truncate-and-force measurement as Causal Importance of Reasoning, $\mathrm{CIR} = \frac{1}{T}\sum_k \mathrm{JS}\big(\mathrm{Bern}(p_k) \,\|\, \mathrm{Bern}(p_T)\big)$ with p_k the probability of the final answer after the first k reasoning tokens, show that outcome rewards lower it on many tasks, and restore it with a CIR bonus in the reward. Our commitment probe is this measurement. What we add is its use *inside the advantage*: the probe's per-segment rise decides which tokens of a step receive that step's credit, in a multi-turn agent, instead of a scalar bonus shared by all tokens. We also check it against resampling and against attention, the two signals Bogdan *et al*. compare.

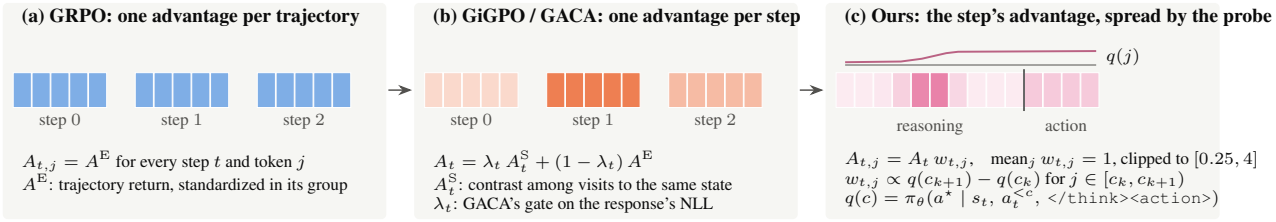


Figure 1. **Credit at three granularities.** GRPO (left) gives every token of a trajectory one episode-level advantage. GiGPO and GACA (middle) give each environment step its own advantage, mixing an anchor-state step contrast A_t^{S} with the episode advantage A^{E} by a weight λ_t. All tokens of the step's response still share it. Probe-weighted credit (right) keeps the step advantage and redistributes it over the response: the commitment probe $q(j)$ scores how likely the eventual action is after the first j tokens, and each segment's credit is proportional to how much q rises across it (weights clipped to $[0.25, 4]$ and renormalized to mean one).

3. Background: Credit in Critic-Free Agent RL

Setting. An agent interacts with an environment for up to T steps. At step t the policy π_θ reads a prompt built from the task, recent history and the admissible actions, and emits a response $a_t = (a_t^1, \ldots, a_t^{|a_t|})$: reasoning inside <think> tags followed by a command inside <action> tags. The environment executes the command; the episode reward is sparse, $r_T \in \{0, 1\}$ for success, plus a small penalty for malformed responses [5]. For each task a group of G trajectories $\{\tau_i\}_{i=1}^{G}$ is sampled with the same initial state.

Clipped surrogate. All methods below share PPO's token-level objective [18]

$$\mathcal{J}(\theta) = \mathbb{E}\Big[\tfrac{1}{|a_t|}\textstyle\sum_j \min\big(\rho_{t,j} A_{t,j},\ \mathrm{clip}(\rho_{t,j}, 1\pm\epsilon)\, A_{t,j}\big)\Big] - \beta\, \mathrm{KL}(\pi_\theta \,\|\, \pi_{\mathrm{ref}}), \tag{1}$$

with $\rho_{t,j}$ the token probability ratio to the sampling policy. They differ only in the advantage $A_{t,j}$, and until now $A_{t,j}$ has not depended on j.

GRPO: episode credit. GRPO [19] standardizes each trajectory's return within its group, $A_i^{\mathrm{E}} = (R(\tau_i) - \mu_G)/(\sigma_G + \varepsilon)$, and broadcasts it to every token of every step.

GiGPO: step credit from anchor states. GiGPO [5] collects every visit, across the group's rollouts, to the same environment state z (identical observation), forming an anchor group $\mathcal{G}_z$ of size m_z. With the discounted step return $\hat{g}_t = \sum_{t' \geq t} \gamma^{t'-t} r_{t'}$, the step advantage standardizes $\hat{g}_t$ within the anchor group, $A_t^{\mathrm{S}} = (\hat{g}_t - \mu_{\mathcal{G}_z})/(\sigma_{\mathcal{G}_z} + \varepsilon)$ (falling back to prompt-group statistics when $m_z = 1$ [11]), and the two levels are summed, $A_t = A^{\mathrm{E}} + w\, A_t^{\mathrm{S}}$.

GACA: state-adaptive mixing. GACA [11] replaces the sum by a per-step convex mixture

$$A_t = \lambda_t\, w\, A_t^{\mathrm{S}} + (1 - \lambda_t)\, A^{\mathrm{E}}. \tag{2}$$

The weight comes from a criticality score: the mean token NLL of the sampled response, $H_t = -\frac{1}{|a_t|}\sum_j \log \pi_{\theta_{\mathrm{old}}}(a_t^j \mid s_t, a_t^{<j})$, normalized by its trajectory mean, $c_t = H_t/\bar{H}$. A piecewise-linear gate maps it to the weight, $\lambda_t = \lambda_0(1 + \alpha(c_t - 1))$ for $c_t \geq 1$ and $\lambda_0(1 - \beta(1 - c_t))$ otherwise, clamped to $[0, 1]$, with $\lambda_0 = 0.5$ and $\alpha = \beta = 0.5$. Uncertain steps lean on the step contrast, confident steps on the episode return. GACA's analysis gives the MSE-optimal weight as $\lambda_z^\star = M_E/(M_E + M_S)$, the ratio of the two estimators' errors, and shows it is non-decreasing in m_z; the NLL enters as a one-sided certificate of local decision criticality. In every case above, all $|a_t|$ tokens of a response share A_t.

4. Probe-Weighted Token Credit

The commitment probe. Let $a^\star$ be the command the response eventually issued and c a cut point in the response. The probe asks how strongly the reasoning written so far commits the model to $a^\star$:

$$q(c) = \pi_\theta\big(a^\star \,\big|\, s_t,\ a_t^{<c},\ \texttt{</think><action>}\big), \tag{3}$$

the probability of the original command's tokens when the reasoning is cut after c tokens and closed by force. It needs one teacher-forced forward pass per cut and no sampling. This is the quantity p_k of CIR [28] and the early-answering test of [10]; $q(0)$ is the probability of the eventual action before any reasoning, and q reaches one by construction once the command is written.

From probe to token weights. We place cuts $0 = c_0 < c_1 < \cdots < c_K$ every 16 reasoning tokens and at the start of the <action> tag, and treat the response end L as a final cut with $q = 1$. Segment k, the tokens $[c_k, c_{k+1})$, receives the rise of the probe across it, spread evenly over its tokens:

$$\tilde{w}_j = \frac{\max\big(q(c_{k+1}) - q(c_k),\, 0\big)}{c_{k+1} - c_k}, \qquad j \in [c_k, c_{k+1}). \tag{4}$$

The raw weights are normalized to mean one over the response, clipped to $[0.25, 4]$ and renormalized, iterating to a fixed point. The step's token advantages are then

$$A_{t,j} = A_t \cdot w_{t,j}, \qquad \tfrac{1}{|a_t|}\textstyle\sum_j w_{t,j} = 1, \tag{5}$$

with A_t from GACA (Eq. (2)). Because the weights average one, the length-normalized contribution of each step in Eq. (1) is unchanged; only where the credit lands inside the response moves. A flat probe (the action was certain from the start) gives uniform weights, recovering GACA exactly, and malformed responses with no action keep uniform weights.

What it does and does not change. Rollouts, the importance ratio and the KL term are untouched, so training stays on-policy; the probe only decides how much each token's gradient counts, much as a fixed reward model would. It says *where* a step's decision was made, not whether the decision was good, which remains the job of A_t. A good step therefore reinforces mostly the reasoning that produced its action, and a bad step penalizes mostly that reasoning rather than the surrounding boilerplate. Under the assumption that a step's value flows through its action, which holds in ALFWorld where only the command touches the environment, $A_t w_{t,j}$ approximates each token's causal contribution to the step's value. The method is orthogonal to the step estimator and applies to GRPO or GiGPO advantages unchanged.

Cost. With cuts every 16 tokens a ∼95-token response needs about seven scoring passes over prompts of ∼500 tokens, computed in the same batched routine that recomputes the old-policy log-probabilities. In our runs this added about 150 s to a ∼390 s iteration (Sec. 7); generation is unaffected and inference uses no probe at all.

Variants. We test three ways of applying the probe, each motivated by a failure mode of concentrating credit: *(i)* a tighter clip $[0.5, 2]$, in case large weights make updates too big; *(ii)* probe weights only where $A_t > 0$, in case a wrong-signed advantage concentrated on the decisive reasoning is especially harmful at low success rates; and *(iii)* scoring the probe with the frozen reference policy instead of the current one, to break a possible feedback loop in which the policy grades its own reasoning and credit for early commitment pushes the decision earlier, the pattern [28] report for outcome rewards. We also evaluate two step-level changes to GACA that target its estimator's reliability rather than tokens: scaling the gate's odds by $1 - 1/m_z$ (GACA's own analysis says $\lambda^\star$ grows with m_z, but m_z never enters its gate; a singleton anchor group gets $\lambda_t = 0$) and the leave-one-out score of GACA's Appendix B.4.

	GACA [11]	Ours
Backbone	Qwen2.5-1.5B-Instr.	same, full FT
Response format	think + action, 512 tok.	same
Max env. steps / group size	50 / 8	50 / 8
lr / KL / γ / penalty	10^{-6} / 0.01 / 0.95 / 0.1	same
Tasks per iteration	16	4
Iterations	150	34
Trajectories per method	19,200	1,088
Seeds per method	3	1
Held-out episodes	128	128 (shared)
Reported checkpoint	best	final

Table 1. **Training configuration.** Everything except the budget follows GACA's shared configuration; every method in our study differs only in the advantage estimator.

5. Experimental Setup

Benchmark and model. We use ALFWorld [21], the text household benchmark on which GACA reports its ablations, with all six task types (pick-and-place, examine in light, clean, heat, cool, and pick two) and a 50-step horizon. The policy is Qwen2.5-1.5B-Instruct [16], the paper's smaller scale, fully fine-tuned. Training runs in verl-agent [5, 20], the stack GACA builds on, with the authors' ALFWorld prompt (ReAct-style [26] reasoning in `<think>` tags, a command in `<action>` tags, 512-token responses, a two-step history) and vLLM [9] rollouts. Each method is a different advantage estimator plugged into the same trainer; GRPO, GACA and all treatments call one shared, unit-tested implementation.

Recipe and budget. We keep GACA's shared configuration (Tab. 1): group size 8, learning rate 10^{-6}, $\gamma = 0.95$, KL coefficient 0.01 with the low-variance estimator, invalid-format penalty 0.1, mean-and-std normalization, $\lambda_0 = 0.5$, $\alpha = \beta = 0.5$. The budget is the binding difference. GACA reports 371 s per iteration on hardware with 62 GB of memory per GPU and trains 150 iterations of 16 tasks × 8 rollouts. On one A100 40 GB our rollout speed matches theirs (228 to 334 s), but the training passes are an order of magnitude slower, so a 16-task iteration costs 1,076 s. Within a budget of about four hours of training per method we use 4 tasks × 8 rollouts and a fixed 34 iterations: 1,088 trajectories per method, about 6% of the paper's 19,200. All curves are still rising at the end; absolute success rates are therefore far below the paper's, and the comparison is of early learning at matched rollout budget.

Evaluation. Every method is evaluated every 10 iterations and at the end on the *same* 128 held-out games (ALFWorld's in-distribution validation split, one episode each, sampling temperature 0.4 as in the authors' recipe). Because the games are shared, two methods are compared game by game with an exact McNemar test [13] on the games one method solves

and the other does not, which is far more sensitive than comparing two rates at $n = 128$. We report the final checkpoint; the paper's best-checkpoint convention picks the maximum of several noisy evaluations and is optimistic. One game is 0.78 points. The base model solves 6 of 128 (4.7%).

Methods. The controls are GRPO and GACA. The treatments are GACA with the m_z reliability gate, GACA with the leave-one-out score (GACA-LOO), GACA with probe-weighted token credit, and the probe's three variants (Sec. 4). One seed per method; the probe arm used a different training-environment seed (the order in which training games are drawn) than the other arms, with the evaluation games pinned to be identical. Earlier in this project, two seeds of the same method differed by 3.1 points at a matched checkpoint, so differences of a few points between single runs are within seed noise.

6. Where Is the Decision Made?

Before training, we measure. All studies in this section use no training and run on real steps sampled by the policy in the format above.

6.1. GACA's step score is mostly path noise

GACA's gate reads the mean NLL of the one response that was sampled. Exact token entropy, $-\sum_v \pi(v) \log \pi(v)$ over the vocabulary, is available from the same forward pass and does not depend on which token was drawn, so it removes the single-sample noise of the NLL at each position. On 7,864 step rows from the first five iterations of a GACA-family run, the two scores agree almost perfectly (Pearson 0.94, within-trajectory Spearman 0.90): replacing NLL by exact entropy moves λ_t by more than 0.1 on only 0.9% of steps. Averaging over ∼95 tokens already cancels token-level sampling noise. The noise that remains is of another kind. At anchor states visited at least twice in a group (575 states), visits to the *same* state differ only in which reasoning path was sampled, and that within-state variation accounts for 87% of the variance of the NLL score (85% for exact entropy). GACA's criticality score is therefore mostly a property of the sampled path, not of the state whose criticality it is meant to certify, which may explain why the paper's own random-permutation control is needed to show that the gate's placement matters. Averaging across visits (GACA-LOO) removes this noise but also the per-visit contrast; we test it in Sec. 7.

6.2. A counterfactual flip test

To find where a decision is made inside a response, we intervene on it. For a step with response of L tokens and issued command $a^\star$, we cut the response after j tokens at about 18 points (every $L/16$ tokens plus the `<action>` tag), resample the remainder 8 times at temperature 1, and record $F(j)$, the fraction of continuations whose command differs from $a^\star$. F falls from $F(0)$ (resample everything) to $F(L) = 0$, and its drop across a segment, $F(c_k) - F(c_{k+1})$, is the share of the decision settled inside it. Because ALFWorld is deterministic, an unchanged command implies an identical next state, so this is an exact step-level relevance measure. We sample 320 valid steps from base-policy rollouts on 48 training games, which costs 44k continuations and about five minutes on one GPU with vLLM. The study is run twice on disjoint game sets: once to design weighting schemes and once held out to test them.

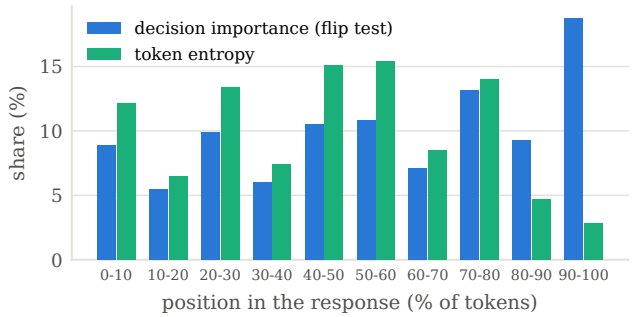


Figure 2. **Where the decision is made vs. where the entropy is**, by position in the response (held-out flip test, 320 steps). The final tenth of a response holds 18.8% of the decision but 2.8% of the token entropy.

Decisions are genuinely open and are made in the reasoning. Resampling a whole response changes the command 72% of the time, and 82% of the decision mass is settled before the `<action>` tag. Figure 2 shows where. Decision importance is spread through the reasoning and peaks in its last tenth, where the model commits (18.8%). Entropy follows a different profile: it is high in the middle of the reasoning and lowest at the end (2.8%), because once the model has committed, the tokens that express the commitment look certain.

6.3. Which signal locates the decision?

We score each token-weighting scheme by the share of its credit that lands on decision importance, $\sum_k m_k\, d_k$ for credit shares m_k and importance shares d_k over the fine flip-test segments, relative to uniform credit (GACA), and by total-variation distance to importance. Figure 3 summarizes. Token entropy, the signal behind single-turn "forking token" methods [24], moves credit toward decisive segments by only 2–4% and fits importance *worse* than uniform (TV $+0.011$, 95% CI $[+0.003, +0.018]$). Sampled NLL does no better, and a fixed $2\times$ weight on the action tokens gives 4–5%. A hybrid designed on the first run (entropy on the reasoning, protected action tokens) did not hold out of sample ($1.07\times$). The commitment probe is the only scheme that fits importance better than uniform: its unclipped segment

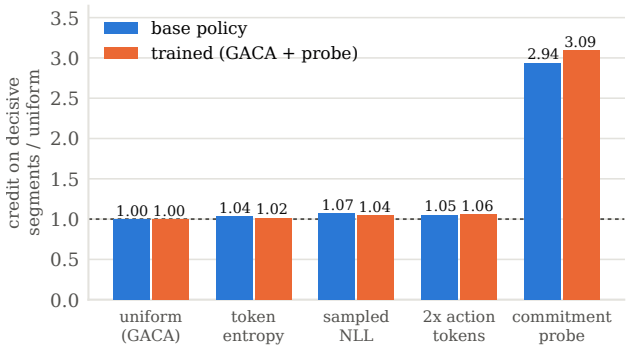


Figure 3. **Credit on decisive segments relative to uniform credit**, by weighting scheme, on held-out games, for the base policy and for the final policy of the GACA + probe training run. Unclipped segment weights; the clipped training configuration gives $1.65\times$ and $1.64\times$.

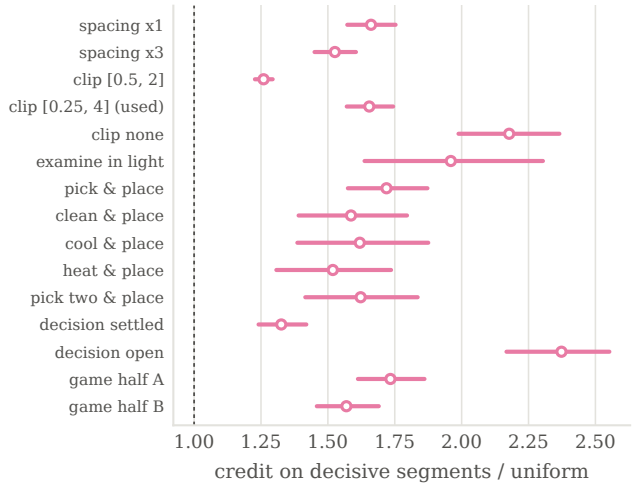


Figure 4. **Robustness of probe weights** at the training configuration: credit on decisive segments relative to uniform with 95% bootstrap intervals, by cut spacing, clip, task type, decision openness and game half.

weights put $2.94\times$ more credit on decisive segments (TV -0.065, CI $[-0.093, -0.037]$).

Robustness at the training configuration. Figure 4 evaluates the exact function used in training (clip $[0.25, 4]$, mean-preserving). It gives $1.65\times$ (CI $[1.57, 1.74]$) and beats uniform on 80% of steps. The effect holds for every task type (1.52 to $1.96\times$), for both halves of the games (1.73 and 1.57), for early and late steps, and with cuts up to three times coarser (1.53). It is largest where it matters most, on steps whose decision was open ($2.37\times$ against $1.33\times$ when settled). The clip trades concentration for stability: unclipped weights give $2.18\times$, a $[0.5, 2]$ clip $1.26\times$.

6.4. Does attention locate it?

Bogdan *et al*. [3] use attention between sentences as a white-box alternative to resampling. A transformer computes, for each query token, a softmax-normalized distribution over earlier keys, $\mathrm{softmax}(QK^\top/\sqrt{d})$ [22]. We read the attention that the command tokens pay to each reasoning token, averaged over heads and over the last third of the layers, and score it like a weighting scheme on 128 held-out steps, restricted to the reasoning segments (where attention is defined) and compared with the probe on the same segments. Attention does point toward the decision, but weakly: it places $1.12\times$ more credit on decisive segments than uniform (CI $[1.03, 1.24]$), against $1.61\times$ ($[1.45, 1.77]$) for the probe on the same steps, and its per-step rank correlation with importance density is only 0.14. Figure 5 shows why on one step. More than half of the decision is settled in the single segment "*[let's check] the drawers to see if any have*", which commits the agent to searching the drawers before it writes `go to drawer 1`. The probe puts its weight there. Attention is diffuse and peaks on the tokens just before the action tag, which is consistent with the view that attention weights are not, by themselves, explanations [7].

6.5. The probe survives training

The flip test above uses the base policy. Repeating it on the same held-out games with the final weights of our GACA + probe training run gives the same picture: the probe's credit on decisive segments is $1.64\times$ at the training configuration (CI $[1.55, 1.72]$; base model $1.65\times$) and $3.09\times$ unclipped, entropy stays near $1\times$, and the share of the decision inside the action tags moves only from 18% to 21%. Training does not degrade the probe as a measurement of where decisions are made.

7. Training Results

Table 2 lists the final held-out success of every method together with paired comparisons against both controls, and Fig. 6 shows the learning curves.

Controls. Both controls learn: GRPO reaches 14.1% and GACA 18.8% from the base model's 4.7% (both $p < 0.05$ against the base model, game-paired). GACA leads GRPO, as the paper reports, but by 13 games to 7 ($p = 0.26$): at 6% of the paper's budget the ordering is the expected one and not resolved. Under the paper's response format our GACA diagnostics match its Table 7 closely: criticality standard deviation 0.35 (paper 0.31), λ_t standard deviation 0.087 (0.08), anchor coverage 0.96 (0.86). This indicates the estimator is reproduced faithfully even though the training budget is not.

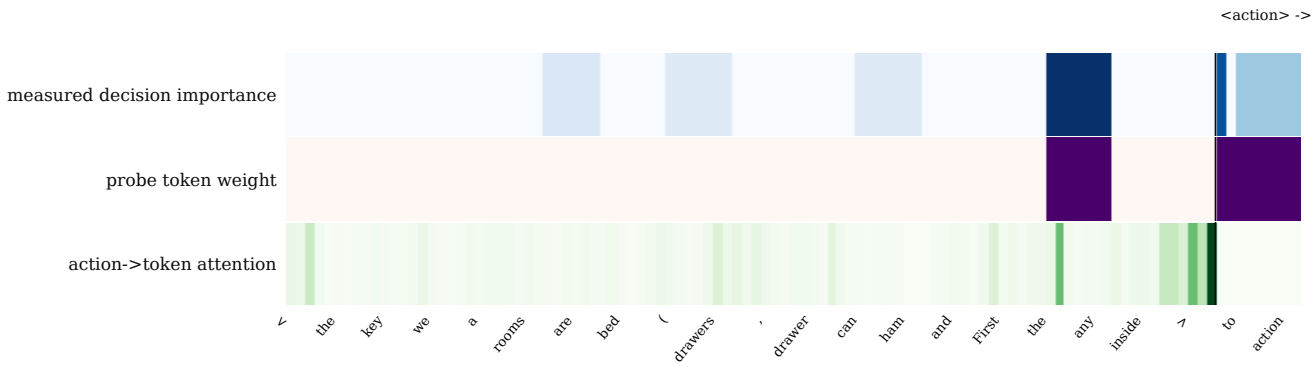


Figure 5. **One held-out step, token by token.** Top: decision importance measured by the flip test (darker = more of the decision settled there). Middle: probe-weighted credit. Bottom: attention from the command tokens to each reasoning token (last third of the layers, mean over heads). The vertical line marks the <action> tag. The task is to put a keychain on a desk; before any reasoning the eventual command has probability $q(0) = 0.04$ and resampling the whole response changes it 75% of the time. 53% of the decision is settled in one segment, "the drawers to see if any have", where the probe also concentrates its weight; attention is spread out and peaks next to the action tag.

Method	Final	Best	vs. GRPO	vs. GACA
Base model (no training)	4.7	–	–	–
GRPO [19]	14.1	14.1	–	13/7 (0.26)
GACA [11]	18.8	18.8	7/13 (0.26)	–
GACA + m_z gate	12.5	12.5	13/11 (0.84)	17/9 (0.17)
GACA-LOO (App. B.4 of [11])	11.7	11.7	11/8 (0.65)	14/5 (0.064)
GACA + probe (ours)	20.3	20.3	8/16 (0.15)	14/16 (0.86)
clip $[0.5, 2]$	14.1	14.1	13/13 (1.00)	16/10 (0.33)
positive A_t only	5.5	6.2	**16/5 (0.027)**	**20/3 (¡0.001)**

Table 2. **Held-out success (%) on the same 128 ALFWorld games**, 34 training iterations (33 for the two GACA step variants), one seed per method. *Final* is the end-of-training checkpoint, *Best* the maximum over five evaluations. The comparison columns give b/c (McNemar p): b games solved only by the control, c only by the method in that row; bold $p < 0.05$.

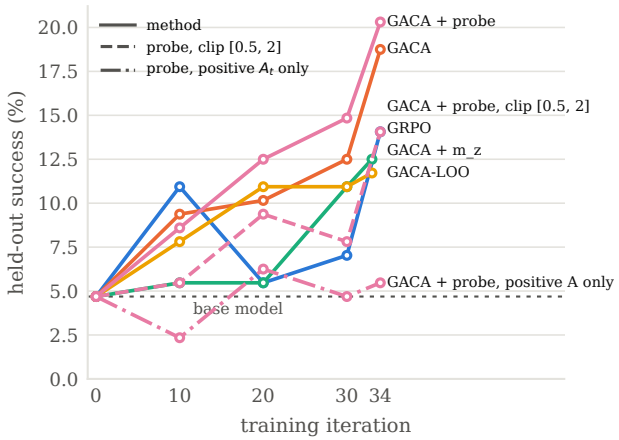


Figure 6. **Held-out success over training** (128 games, evaluated every 10 iterations). Colour follows the method; variants of the probe keep its colour and change line style.

Probe-weighted token credit. GACA + probe reaches 20.3%, the highest final success of any method, after rising at every evaluation (4.7, 8.6, 12.5, 14.8, 20.3) and leading every arm from iteration 20 on. It also has the highest training-batch success (11.2% averaged over iterations, against 9.2% for GACA). Game by game it solves 16 games GACA misses while missing 14 that GACA solves ($p = 0.86$), and against GRPO the split is 16 to 8 ($p = 0.15$). We read this as *consistent with* a benefit, not as evidence for one: with one seed per arm and a 3-point seed-to-seed spread observed earlier, a 1.5-point lead over GACA is well within noise. The probe's cost is visible in wall-clock: 550 s per iteration against 389 s for GACA, of which 151 s is the probe passes. Its updates are moderately larger than GACA's (mean gradient norm 1.06 against 0.84, KL 0.024 against 0.020), as expected when the same advantage is concentrated on fewer tokens. The probability of the eventual action before any reasoning, $q(0)$, *fell* from 0.19 to 0.14 over training, so the reasoning became more causally important rather than decorative, the opposite of the drift Yu *et al.* [28] warn about for outcome rewards.

How the probe is applied matters. The three variants were designed to fix failure modes of concentration, and none helped. A tighter clip $[0.5, 2]$ brought the gradient norm to GACA's level (0.93) and finished at 14.1%, level with GRPO. Applying the probe only to positive advantages, so that blame always stays uniform, was the worst method: 5.5%, essentially no learning, and significantly below both GRPO (16 to 5, $p = 0.027$) and GACA (20 to 3, $p < 0.001$). Its KL rose to 0.059 over the last iterations although $q(0)$ stayed flat (0.20 to 0.19), so early commitment does not explain the failure. A plausible reading is that the asymmetry itself is harmful: positive steps push hard on a few decisive

tokens while negative steps push weakly on all tokens, so the policy is moved toward specific reasoning patterns without a matching signal moving it away from bad ones. **[TODO: frozen-reference arm result and reading.]**

Step-level treatments. Neither change to GACA's step estimator helped. Scaling the gate's odds by $1 - 1/m_z$ changes little because 96% of steps already fall in anchor groups with $m_z \geq 2$; it finished at 12.5%. The leave-one-out score of GACA's Appendix B.4, which averages away the within-state noise we measured, finished at 11.7% and trails GACA by 14 games to 5 ($p = 0.064$). Pooling the score also shrinks the spread of λ_t to 0.039 from 0.087, so the gate barely adapts; in the paper's terms, it trades the noise of the single-sample score for the loss of within-group resolution, and at this budget the trade does not pay.

8. Discussion and Limitations

What the evidence supports. The measurement results are the firm part of this paper. They are counterfactual, repeated on disjoint game sets and on a trained policy, and come with intervals. Inside an agent's reasoning the decision is made before the command is written; uncertainty signals (entropy, NLL) do not locate it; the commitment probe does. GACA's criticality score is dominated by within-state path noise. The training results are weaker. Probe-weighted credit had the highest final success, and the variants that changed how it is applied did worse, two of them significantly. With one seed per arm and about 6% of the published training budget, no difference between the main methods is statistically resolved.

Why a better measurement need not train better. The probe changes where credit lands, not whether it is right. At 5–15% training success most step advantages rest on one or two successful rollouts in a group of eight, so their sign is often wrong; concentrating a wrong-signed advantage on the reasoning that produced the decision does more damage than spreading it. Uniform credit is, in this sense, a hedge against a noisy step estimator, and the positive- only variant shows that asymmetric hedging is worse than none. We expect probe weighting to pay off more where step advantages are reliable: at higher success rates, with more rollouts per group, or combined with an estimator that scales concentration by its own confidence, for instance the number of successful rollouts in the group or whether the anchor group carries a real return contrast.

Limitations. *Budget and seeds*: about 6% of GACA's trajectories and one seed per method, so method differences of a few points are not interpretable on their own; the paired tests are the right lens, and they are mostly inconclusive. *Training seed*: the probe arm drew its training games in a different order than the controls, with identical evaluation games. *Benchmark*: ALFWorld only; WebShop, the paper's second benchmark, needs a product-search server we did not stand up. *Oracle*: the flip test measures relevance for the next command only, not for the episode's outcome, and uses 8 resamples per cut; importance is a difference of two noisy rates. *Probe*: as a likelihood ratio under teacher forcing, the probe can credit a segment that makes the eventual command *expressible* rather than one that decides it. *Cost*: the probe adds about 40% to an iteration in our setup; computing it from a prefix-shared batch would reduce this.

Outlook. The cheapest next steps follow directly: multiple seeds of the probe and GACA at this budget, the paper's 16-task batch, and probe weights scaled by a confidence in the step advantage. More broadly, our results suggest a separation of concerns for critic-free agent RL: the step estimator decides *how much* credit a step earns, and a counterfactual probe decides *which tokens* earned it. The second is now cheap and measurable; the first is the bottleneck.

9. Conclusion

We asked where, inside an agent's reasoning, the decision is made. A counterfactual flip test answers that it is made in the reasoning, mostly not where the model is uncertain, and that a teacher-forced commitment probe, the measurement behind CIR, finds it reliably. We turned the probe into a mean-preserving redistribution of GACA's step advantage over tokens. In a budget-matched, game-paired study on ALFWorld it achieved the best final success of the methods we trained, but the margins are within single-seed noise, and ways of applying it that seemed safer did worse. Measuring where credit belongs has become easy; making the credit itself reliable is the open problem.

References

[1] Arash Ahmadian, Chris Cremer, Matthias Gallé, Marzieh Fadaee, Julia Kreutzer, Olivier Pietquin, Ahmet Üstün, and Sara Hooker. Back to basics: Revisiting REINFORCE-style optimization for learning from human feedback in LLMs. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics*, pages 12248–12267, 2024. 2

[2] Jose A. Arjona-Medina, Michael Gillhofer, Michael Widrich, Thomas Unterthiner, Johannes Brandstetter, and Sepp Hochreiter. RUDDER: Return decomposition for delayed rewards. In *Advances in Neural Information Processing Systems*, 2019. 2

[3] Paul C. Bogdan, Uzay Macar, Neel Nanda, and Arthur Conmy. Thought anchors: Which LLM reasoning steps matter? *arXiv preprint arXiv:2506.19143*, 2025. 2, 6

[4] Ganqu Cui, Yuchen Zhang, Jiacheng Chen, Lifan Yuan, Zhi Wang, Yuxin Zuo, Haozhan Li, Yuchen Fan, Huayu Chen,

Weize Chen, et al. The entropy mechanism of reinforcement learning for reasoning language models. *arXiv preprint arXiv:2505.22617*, 2025. 2

[5] Lang Feng, Zhenghai Xue, Tingcong Liu, and Bo An. Group-in-group policy optimization for LLM agent training. In *Advances in Neural Information Processing Systems*, 2025. 1, 2, 3, 4

[6] Daya Guo, Dejian Yang, Haowei Zhang, Junxiao Song, Peiyi Wang, Qihao Zhu, Runxin Xu, Ruoyu Zhang, Shirong Ma, Xiao Bi, et al. DeepSeek-R1 incentivizes reasoning in LLMs through reinforcement learning. *Nature*, 645(8081):633–638, 2025. 1, 2

[7] Sarthak Jain and Byron C. Wallace. Attention is not explanation. In *Proceedings of NAACL-HLT*, pages 3543–3556, 2019. 6

[8] Amirhossein Kazemnejad, Milad Aghajohari, Eva Portelance, Alessandro Sordoni, Siva Reddy, Aaron Courville, and Nicolas Le Roux. VinePPO: Unlocking RL potential for LLM reasoning through refined credit assignment. *arXiv preprint arXiv:2410.01679*, 2024. 2

[9] Woosuk Kwon, Zhuohan Li, Siyuan Zhuang, Ying Sheng, Lianmin Zheng, Cody Hao Yu, Joseph E. Gonzalez, Hao Zhang, and Ion Stoica. Efficient memory management for large language model serving with PagedAttention. In *Proceedings of the 29th Symposium on Operating Systems Principles*, 2023. 4

[10] Tamera Lanham, Anna Chen, Ansh Radhakrishnan, Benoit Steiner, Carson Denison, Danny Hernandez, Dustin Li, Esin Durmus, Evan Hubinger, Jackson Kernion, et al. Measuring faithfulness in chain-of-thought reasoning. *arXiv preprint arXiv:2307.13702*, 2023. 2, 3

[11] Taoran Liang, Yang Liu, Shang Luo, Yingguang Yang, Rongrong Zhang, Yingzong Min, Yulin Huang, Jianshen Zhang, Yongzhi Qi, Kefu Xu, Congjing Ran, and Bin Chong. Granularity-adaptive credit assignment for long-horizon LLM agent reinforcement learning. In *International Conference on Learning Representations*, 2026. 1, 2, 3, 4, 7

[12] Hunter Lightman, Vineet Kosaraju, Yuri Burda, Harrison Edwards, Bowen Baker, Teddy Lee, Jan Leike, John Schulman, Ilya Sutskever, and Karl Cobbe. Let's verify step by step. In *International Conference on Learning Representations*, 2024. 2

[13] Quinn McNemar. Note on the sampling error of the difference between correlated proportions or percentages. *Psychometrika*, 12(2):153–157, 1947. 4

[14] Andrew Y. Ng, Daishi Harada, and Stuart Russell. Policy invariance under reward transformations: Theory and application to reward shaping. In *International Conference on Machine Learning*, pages 278–287, 1999. 2

[15] Long Ouyang, Jeffrey Wu, Xu Jiang, Diogo Almeida, Carroll Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, et al. Training language models to follow instructions with human feedback. *Advances in Neural Information Processing Systems*, 35:27730–27744, 2022. 2

[16] Qwen Team. Qwen2.5 technical report. *arXiv preprint arXiv:2412.15115*, 2024. 4

[17] John Schulman, Philipp Moritz, Sergey Levine, Michael Jordan, and Pieter Abbeel. High-dimensional continuous control using generalized advantage estimation. *arXiv preprint arXiv:1506.02438*, 2015. 2

[18] John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. Proximal policy optimization algorithms. *arXiv preprint arXiv:1707.06347*, 2017. 2, 3

[19] Zhihong Shao, Peiyi Wang, Qihao Zhu, Runxin Xu, Junxiao Song, Xiao Bi, Haowei Zhang, Mingchuan Zhang, Y. K. Li, Yang Wu, and Daya Guo. DeepSeekMath: Pushing the limits of mathematical reasoning in open language models. *arXiv preprint arXiv:2402.03300*, 2024. 1, 2, 3, 7

[20] Guangming Sheng, Chi Zhang, Zilingfeng Ye, Xibin Wu, Wang Zhang, Ru Zhang, Yanghua Peng, Haibin Lin, and Chuan Wu. HybridFlow: A flexible and efficient RLHF framework. In *Proceedings of the Twentieth European Conference on Computer Systems*, pages 1279–1297, 2025. 2, 4

[21] Mohit Shridhar, Xingdi Yuan, Marc-Alexandre Côté, Yonatan Bisk, Adam Trischler, and Matthew Hausknecht. ALFWorld: Aligning text and embodied environments for interactive learning. *arXiv preprint arXiv:2010.03768*, 2020. 4

[22] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. *Advances in Neural Information Processing Systems*, 30, 2017. 6

[23] Peiyi Wang, Lei Li, Zhihong Shao, Runxin Xu, Damai Dai, Yifei Li, Deli Chen, Yu Wu, and Zhifang Sui. Math-shepherd: Verify and reinforce LLMs step-by-step without human annotations. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics*, pages 9426–9439, 2024. 2

[24] Shenzhi Wang, Le Yu, Chang Gao, Chujie Zheng, Shixuan Liu, Rui Lu, Kai Dang, Xiong-Hui Chen, Jianxin Yang, Zhenru Zhang, et al. Beyond the 80/20 rule: High-entropy minority tokens drive effective reinforcement learning for LLM reasoning. In *Advances in Neural Information Processing Systems*, 2025. 1, 2, 5

[25] Zihan Wang, Kangrui Wang, Qineng Wang, Pingyue Zhang, Linjie Li, Zhengyuan Yang, Xing Jin, Kefan Yu, Minh Nhat Nguyen, Licheng Liu, et al. RAGEN: Understanding self-evolution in LLM agents via multi-turn reinforcement learning. *arXiv preprint arXiv:2504.20073*, 2025. 1, 2

[26] Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan Cao. ReAct: Synergizing reasoning and acting in language models. *arXiv preprint arXiv:2210.03629*, 2022. 4

[27] Qiying Yu, Zheng Zhang, Ruofei Zhu, Yufeng Yuan, Xiaochen Zuo, Yu Yue, Weinan Dai, Tiantian Fan, Gaohong Liu, Lingjun Liu, et al. DAPO: An open-source LLM reinforcement learning system at scale. In *Advances in Neural Information Processing Systems*, 2025. 2

[28] Qinan Yu, Alexa Tartaglini, Peter Hase, Carlos Guestrin, and Christopher Potts. Outcome rewards do not guarantee verifiable or causally important reasoning. *arXiv preprint arXiv:2604.22074*, 2026. ICML 2026. 2, 3, 4, 7

[29] Yifei Zhou, Andrea Zanette, Jiayi Pan, Sergey Levine, and Aviral Kumar. ArCHer: Training language model agents via

hierarchical multi-turn RL. *arXiv preprint arXiv:2402.19446*, 2024. 2